

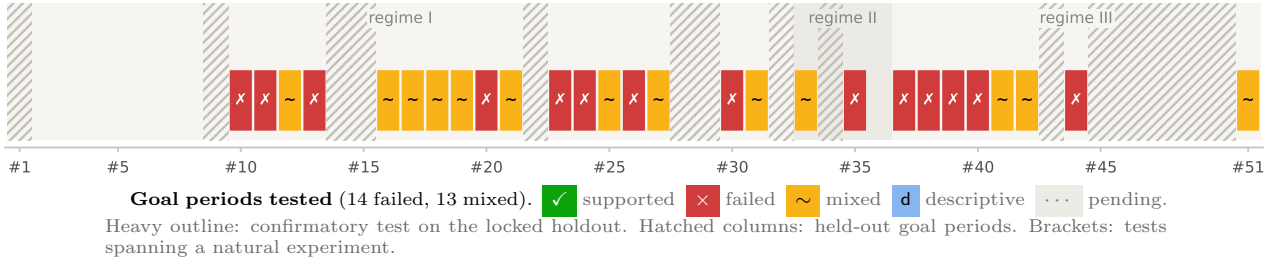
# H17: Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

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**Bottom line.** Behavior on action classes is a non-Markov mix of tool modes and idling. Its slowest timescale is a stable but uninformative period statistic (not an order parameter for stuckness), and the content-level test awaits the Jev states.

level: hypothesis · pipeline checks 100% · updated 2026-10-03



## WHAT HAPPENED

Is agent behavior a Markov state model (MSM) with a few slow metastable sets, and is its slowest timescale  $t_2^*$  a per-period order parameter, slowest in stuck periods? Round 1 (exploratory) fitted MSMs on 6 action classes per active minute in 27 non-holdout periods (17 regime I, 2 regime II, 8 regime III). Synthetic tests showed the estimators recover  $t_2$  and planted sets at village sample sizes. On real data: CK fails in 27/27, and  $t_2(\tau)$  rises  $\times 4-7$  from 1 to 15 min, so the chain is not Markov.  $t_2^*$  (5–24 min) beats a sojourn-preserving null by only  $\times 1.06-1.62$ , which passes in 3/8 regime-III periods. Pooling agents inflates  $t_2^*$  by  $\times 1.28$ .  $t_2^*$  does not track errors or output, regime III is *slower* than regime I (reversed), and a “don’t do nothing” prompt (NE07) lengthened it. Verdicts: 13 mixed, 14 failed. Holdout (#32, #45) not run.

## PHYSICAL MODEL

Kinetic Potts / non-equilibrium Markov chain (models 02, 10). State  $x_t \in \{\text{browse, type, shell, chat, idle, consolidate}\}$  per agent-minute (H14 scheme);  $C(\tau)$ : lagged pair counts within agent-days;  $T(\tau)$ : non-reversible MLE;  $t_k$ : implied timescales from eigenvalues  $\lambda_k$ ;  $\chi$ : PCCA+ memberships (Röblitz-Weber) of  $m$  metastable sets;  $q^\pm$ : forward/backward committers,  $\delta = 0$  under detailed balance;  $p_i$ : soft (Jev) posteriors.

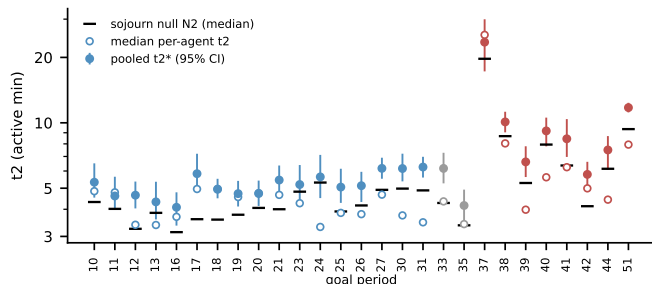
$$T_{ij}(\tau) = C_{ij}(\tau) / \sum_k C_{ik}(\tau), \quad t_k(\tau) = -\tau / \ln |\lambda_k(T(\tau))|$$

$$t_2^* \equiv t_2(\tau_c = 5 \text{ min})$$

$$\text{CK: } P_{A \rightarrow A}(k\tau) \stackrel{?}{=} [\pi_A^\top T(\tau)^k]_A$$

$$T_{\text{cg}} = (\chi^\top \Pi \chi)^{-1} \chi^\top \Pi T \chi, \quad \delta_i = q_i^+ - (1 - q_i^-)$$

$$\lambda_k^\tau = \text{eig}[C(1)^{-1} C(1+\tau)], \quad C(\tau) = \sum_t p_t p_{t+\tau}^\top$$



**Observables.** Slowest implied timescale per goal period at  $\tau_c = 5$  min. Filled: pooled MSM (95% agent-day CI). Bars: sojourn-null median (keeps each state’s dwell times, shuffles their order). Open: median per-agent  $t_2$ . The pooled MSM beats the sojourn null in every period, but only modestly, and it sits above the typical agent (agent-mixture inflation). Regime III (red) is slower than regime I (blue). #37’s peak is agents idling before their first and after their last action of the day.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly: on action classes, this is a curiosity with one methodological payoff. The “metastable sets” are mostly tool modality (GUI vs. shell) and idling, which any log viewer could tell you. Part of the GUI-vs-shell split is GUI-heavy vs. shell-heavy agents.  $t_2^*$  is a stable, cheap period statistic, but it does not detect stuck periods (no link to error share or output), so an operator gains no monitor. The useful part is negative and transferable: minute-scale agent behavior is strongly non-Markov, pooled MSMs over heterogeneous agents manufacture slow modes, and LLM-labelled states with  $\kappa \approx 0.5$  would bias timescales low by 4–14 $\times$  unless a noise-cancelling estimator is used. Whether content-level (Jev) states show real “stuck” attractors is still open.

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